

Explainable Multimodal Vision Transformer Framework for Breast Cancer Diagnosis Using Mammography and Ultrasonography

Abstract. Breast cancer diagnosis in resource-limited settings is often constrained by delayed detection, limited specialist availability, and reliance on individual imaging modalities. While artificial intelligence (AI) has shown strong potential for automated medical image analysis, many existing approaches remain unimodal and lack interpretability, thus limiting clinical trust and practical adoption. In addition, the scarcity of “natively” paired multimodal breast imaging datasets poses a challenge for effective multimodal learning. This paper presents an explainable multimodal vision transformer (**MMIBC**) framework for breast cancer diagnosis that integrates mammography and ultrasonography. The proposed framework employed parallel pretrained Vision Transformer backbones for modality-specific representation learning, followed by feature-level fusion and a lightweight classification head to generate unified diagnostic predictions. To address the absence of paired multimodal datasets, part of the uniqueness of this current study is a label-consistent programmatic pairing and fusion strategy introduced to enable multimodal training under realistic data constraints. Model transparency is enhanced through the integration of Gradient-weighted Class Activation Mapping (Grad-CAM), providing visual explanations that highlight image regions influencing model decisions. Tests conducted using the VinDr-Mammo and Breast Ultrasound Images (BUSI) datasets achieved an overall classification accuracy of 84%. Qualitative explainability results indicate that the model’s predictions are grounded in clinically relevant anatomical regions. Analysis of the results further highlights the influence of dataset imbalance on malignant case detection, underscoring important considerations for real-world deployment. This research work contributed a performance benchmark for multimodal breast cancer diagnosis in resource-limited settings and demonstrated the feasibility of trustworthy, multimodal, and explainable AI for accessible imaging modalities and deployment in low-resource settings.

Keywords: Multimodal Imaging, Breast Cancer Diagnosis, Vision Transformers, Explainable AI, Mammography, Ultrasound.

1 Introduction

Digital Signal Processing (DSP) provides the foundational principles for modern data analysis by using computational algorithms to enhance signal quality and extract

key information [19, 31]. In medical science, DSP is the bedrock of diagnostic imaging, where its techniques are essential for improving image clarity and accuracy for modalities like magnetic Resonance Imaging (MRI), computed tomography (CT) and ultrasound [6]. Medical imaging modalities such as mammography and ultrasonography are, at their core, electromechanical systems that convert physical phenomena, such as X-ray attenuation and high-frequency acoustic wave propagation, into digital signals, the diagnostic value of which is determined by the computational methods applied to them [8, 18, 33]. The diagnostic pipeline spans acquisition, conditioning, feature extraction, and classification, with the possibility of information being lost at each stage [2-3, 36]. Robust digital signal processing (DSP) techniques for noise filtering, contrast enhancement, and feature extraction are therefore fundamental to breast cancer diagnosis rather than supplementary to it [14, 41]. These preprocessing steps are essential for making minor yet important signs of malignancy, such as microcalcification clusters or lesion margins, easier for radiologists to detect when diagnosing breast cancer [18, 41]. Given that breast cancer continues to be one of the world's largest causes of death for women [10], the development of such sophisticated diagnostic technologies is desperately needed.

In places with low medical resources, like Sub-Saharan Africa, where late-stage diagnosis is a major cause of high death rates, the burden of breast cancer is disproportionately high [7]. Challenges such as scarcity of trained radiologists and the high cost of advanced equipment/technology indicate that accessible tools like mammography and ultrasound are the cornerstones of diagnosis [26]. While effective, the diagnostic accuracy of these modalities can be limited when used in isolation, thus creating an urgent need for advanced computational tools that can maximize their potential.

The advent of artificial Intelligence (AI), especially deep learning, has extended the capabilities of DSP by enabling the automated interpretation of complex medical images [42]. Modern AI architectures, particularly Vision Transformers (ViTs), have demonstrated remarkable success in classifying breast cancer images with high accuracy, often rivalling the performance of expert radiologists [20, 25, 32]. However, a major impediment to the clinical adoption of these powerful models is their "black box" nature. Most state-of-the-art systems provide a diagnostic output with little to no insight into their internal reasoning. This lack of transparency erodes clinical trust, as practitioners are understandably reluctant to base critical patient care decisions on technology they cannot understand or verify [17].

To bridge this gap, two parallel research streams have gained prominence. The first is explainable AI (XAI), which focuses on developing techniques like Grad-CAM and SHAP to make model decisions transparent [2]. By generating visual heatmaps or feature attribution scores, XAI provides a crucial mechanism for building the clinical trust necessary for real-world implementation. The second stream is multimodal AI, which seeks to improve diagnostic accuracy by fusing data from multiple sources. Frameworks that combine mammography and ultrasound have shown promise in creating a

more robust and comprehensive diagnostic picture [5]. The clinical justification for this combination is well-established; population-based screening programs have demonstrated that integrating ultrasound with mammography significantly increases cancer detection rates and provides critical complementary structural information, particularly in patients with dense breast tissue [11].

However, despite this advancement, a critical research gap persists at the intersection of these fields. Recent reviews of explainable multimodal approaches for breast cancer diagnosis highlight a growing trend of integrating radiological images with highly complex data types, such as histopathology, genomics, or clinical electronic health records [1]. While these cross-domain approaches achieve impressive predictive performance, they rely on invasive biopsy procedures or advanced laboratory infrastructure that are rarely accessible in low-resource environments like Africa. Furthermore, the explainability techniques applied to such heterogeneous data often yield complex feature attributions that are difficult for frontline clinicians to interpret quickly. Specifically, lack of interpretable multimodal frameworks designed to fuse strictly non-invasive, accessible modalities like mammography and ultrasound data for the unique challenges of resource-limited settings, to the best of the authors knowledge, still exist in the literature. Addressing these gaps is crucial to ensure that both the data requirements and the resulting visual explanations align directly with the realities of constrained clinical workflows.

In this research work, we propose **MMIBC**, an explainable multimodal imaging framework for breast cancer diagnosis that fuses mammography and ultrasound data using a Vision Transformer-based architecture. The framework employs parallel ViT backbones for modality-specific feature extraction, followed by feature-level fusion and an adapted classification head to generate unified diagnostic predictions. To enhance transparency, we integrate Gradient-weighted Class Activation Mapping (Grad-CAM) to provide visual explanations that highlight clinically relevant regions influencing model decisions. Furthermore, a programmatic pairing strategy is introduced to enable multimodal learning from publicly available but unpaired datasets. This approach addressed the research gaps identified in the existing literature: the relatively low number or lack of paired multimodal datasets for cancer diagnosis, limited explainable frameworks, data imbalance effects, and the applicability of multimodal to low-resource settings.

Therefore, the main contributions of this paper are summarized as follows:

- i. An explainable multimodal architecture that utilized publicly available datasets from resource-limited settings with VinDr-Mammo and BUSI (Breast Ultrasound Images) from diverse African and Asian populations, addressing the critical need for context-appropriate diagnostic tools, using state-of-the-art Vision Transformers for robust feature extraction and fusion.
- ii. A practical data pairing strategy that enables baseline multimodal training and evaluation in the absence of natively paired datasets, reflecting real-world constraints in resource-limited environments.

- iii. Transparent diagnostic insights through the integration of Grad-CAM, providing clinically interpretable visual explanations to support trust and validation.
- iv. Context-specific benchmarking, including a detailed evaluation of performance trade-offs and limitations arising from dataset imbalance.

The remainder of this paper is organized as follows. Section 2 covers review of related work on AI-based breast cancer diagnosis, multimodal learning, and explainable systems. Section 3 describes the proposed methodology followed by results and discussion in section 4. Section 5 presents the limitations of the proposed methodology and framework while Section 6 concluded the paper with directions for future research.

2. Review of Related Works

This section dives into the existing approaches that have been used by different researchers for breast cancer diagnosis and their limitations.

2.1 AI in Breast Cancer Diagnosis

The application of artificial intelligence (AI) to breast cancer diagnosis has matured significantly, building upon foundational studies that validated its clinical potential. Large-scale evaluations demonstrated that AI systems could perform breast cancer screening on mammograms at a level comparable to expert radiologists, hence justifying the drive to integrate this technology into clinical workflows [25, 29]. This progress has been accelerated by the development of large-scale, publicly available datasets for both mammography and ultrasound, such as VinDr-Mammo and BUSI, which have become standard benchmarks for training and benchmarking models [4, 29].

The motivation for such technology is particularly strong in resource-limited contexts. Epidemiological studies have detailed the immense challenges of breast cancer diagnosis in Sub-Saharan Africa, where a reliance on accessible modalities like mammography and ultrasound is a clinical necessity due to infrastructural constraints [7, 30]. Within this context, AI-based medical decision support systems are increasingly viewed as a means of augmenting limited clinical expertise rather than replacing it, underscoring the need for approaches that are both reliable and interpretable.

2.2 Vision Transformers in Medical Imaging

As research advances, so is the architectures used to do. Recent studies have shown that Vision Transformers (ViTs) can outperform traditional Convolutional Neural Networks (CNNs) in classifying breast cancer images, primarily due to their superior ability to capture global contextual information [20, 32]. However, the increasing complexity of models like ViTs has amplified the "black box" problem, a major barrier to clinical adoption.

Furthermore, the architecture design of ViTs allows these models to capture long-range dependencies across the entire image because of the ability of the ViTs to divide an image into patches and process them as a sequence. The global perspective is highly advantageous when dealing with complex, heterogeneous tissue structures often used in breast cancer. Specifically, the ability of ViTs to carefully model these global con-

texts makes them exceptionally well-suited for multimodal fusion, as they can effectively integrate diverse data from various imaging sources like mammography and ultrasound [23-24, 28, 40]. By leveraging these self-attention mechanisms, ViTs provide a more critical understanding of breast tissue characteristics than localized CNN filters can achieve alone.

2.3 Explainable AI in Oncology

In response to the “black box” challenge, the field of Explainable AI (XAI) has become a critical area of research. Systematic reviews have highlighted the need for transparent models to build clinician trust, especially in multimodal diagnostics [17]. Consequently, techniques like Gradient-weighted Class Activation Mapping (Grad-CAM) and SHapley Additive exPlanations (SHAP) are now being integrated to provide visual and feature-based justifications for model predictions [2, 30]. Despite their effectiveness, many existing studies incorporate explainability as an auxiliary analysis rather than as a core design consideration.

Beyond mere visualization, recent advancements emphasize the need to rigorously evaluate XAI methods using metrics such as explanation stability and integrity, ensuring that the generated explanations consistently align with the model’s actual decision-making process [21]. To achieve this, researchers are increasingly combining complementary techniques. For instance, while Grad-CAM provides rapid, spatially oriented heatmaps, SHAP offers detailed, feature-level attribution based on game theory, making it easier to quantify the specific factors driving a diagnosis [15, 35]. Additionally, attention-guided mechanisms have proven particularly valuable in multimodal settings by assigning explicit weights to different input regions, thereby providing granular, interpretable outputs that further facilitate clinical validation and trust [9].

2.4 Multimodal Learning for Imaging

Parallel to advances in Explainable AI (XAI), multimodal learning has gained significant traction for improving diagnostic robustness, as traditional unimodal methods often fail to capture cancer’s complex heterogeneity [37-38]. Recent research explores various fusion strategies, ranging from the late fusion of modality-specific features [2, 40] to unified, knowledge-augmented networks capable of handling patient-specific variations [13, 16] to enhance precision oncology [12, 22, 31]. However, integrating multimodal data introduces challenges such as aligning heterogeneous features, managing differing resolutions, and overfitting on limited data [12, 34, 37]. While attention-based mechanisms effectively address these discrepancies by dynamically weighting relevant clinical features [9], their high computational demands often restrict practical application in resource-constrained settings [38, 42]. To overcome these limitations, recent specialized frameworks have proposed novel integration strategies. For instance, Al-Tam et al. [5] developed a hybrid CAD framework that integrates detection and classification using merged, unpaired mammography and ultrasound datasets. Meanwhile, Guo et al. [16] explored multimodal fusion by combining 2-dimensional B-mode ultrasound with Contrast-Enhanced Ultrasound (CEUS) videos using a knowledge-

augmented deep learning approach. However, a significant divergence remains between these studies and our proposed MMIBC framework. Al-Tam et al. [5] combined multiple datasets and applied visual harmonisation techniques to merge data distributions, this work, MMIBC, addressed the lack of paired data with a novel label-consistent programmatic pairing strategy designed for feature-level fusion. Furthermore, unlike Guo et al. [16], which required dynamic video data (CEUS), MMIBC relies strictly on the most widely accessible stable modalities (standard mammography and ultrasonography) tailored for resource-constrained environments.

2.5 Identified Gaps in Prior Studies

While research studies such as Al-Tam et al. [5] and Guo et al. [16] provided an important benchmark by combining multiple imaging modalities within an explainable framework, relatively limited attention has been paid to the practical data constraints that dominate real-world deployment in resource-constrained environments. Challenges such as severe class imbalance and the absence of large, natively paired multimodal datasets are often underemphasized, despite their direct impact on model reliability and generalization. Existing methods often require dynamic imaging like CEUS videos or rely on complex harmonization of different datasets rather than direct feature-level fusion of paired modalities, which is a barrier to future research in low-resource environments. This work addressed these gaps by presenting an interpretable, Vision Transformer-based multimodal framework that explicitly accounts for these constraints through a label-consistent programmatic pairing strategy, positioning the proposed system as a baseline experimental reference for future methodological and architectural improvements.

3. Materials and Method

The methodology employed in this work is structured around the design, implementation, and evaluation of an explainable multimodal framework for breast cancer diagnosis, with an explicit emphasis on establishing a reproducible baseline experimental pipeline suitable for multimodal learning under real-world data constraints. Rather than aggressively optimizing performance, the design choices in this section prioritize interpretability, stability, and extensibility, enabling future methodological and architectural improvements to be evaluated against a clearly defined reference. Figure 1 illustrates the overall pipeline of the proposed framework, from raw input processing to final diagnostic prediction.

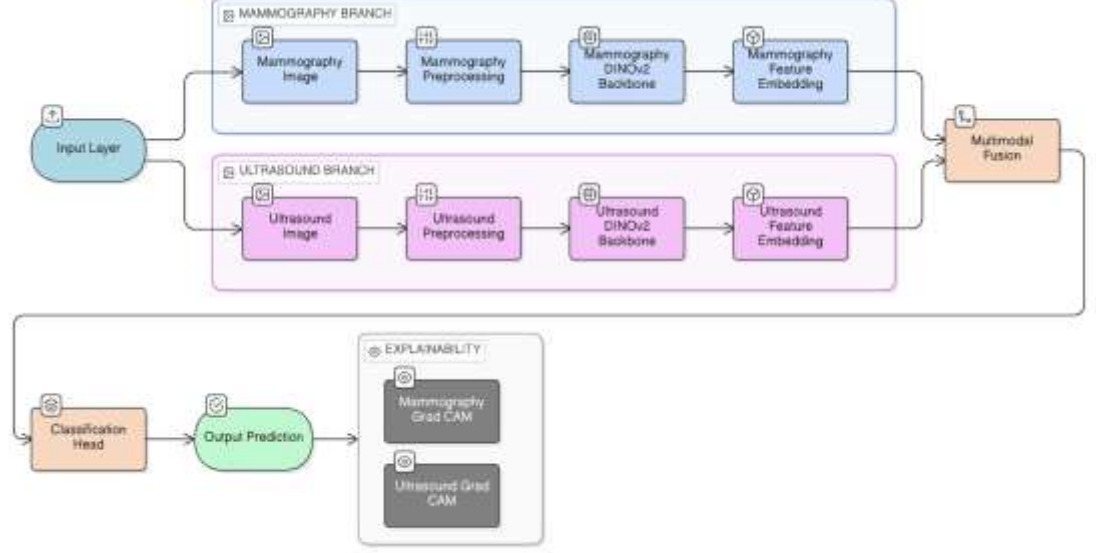


Fig.1. Overall pipeline of the proposed MMIBC framework, illustrating the label-consistent programmatic pairing, dual DINOv2 feature extraction, cross-attention fusion block, and the final classification head.

3.1 Datasets Acquisition

The proposed framework is built upon two publicly available datasets - VinDr-Mammo [29] and Breast Ultrasound Images [4], each representing one of the imaging modalities considered in this study, with comparison with the MMIBC employed in this work presented in table 1.

Table 1: Comparison of Recent Multimodal Approaches for Breast Cancer Diagnosis

Authors	Modalities / Datasets used	Architecture details	Limitations
Al-Tam et al. [5]	Mammography + Ultrasound (Merged Datasets)	YOLOv8 + ViT-based ResNet50	Relies on visual harmonization (t-SNE) for merging disparate datasets.
Guo et al. [16]	B-mode Ultrasound + CEUS Videos	Knowledge-Augmented Deep Learning (KAMnet)	Requires dynamic contrast-enhanced video data, which is less accessible in low-resource settings.
This work (MMIBC)	Mammography + Ultrasound (Unpaired)	Parallel Pre-trained ViTs + Feature-Level Fusion	Programmatic pairing requires careful validation to mitigate synthetic bias.

These datasets differ substantially in scale and acquisition characteristics, reflecting common disparities encountered in real-world multimodal medical imaging research. Sample images from each dataset are shown in Figure 2.

i. VinDr-Mammo serves as the primary source of mammographic data. This large-scale benchmark dataset, available through PhysioNet [29], contains 5,000 four-view full-field digital mammography examinations annotated by expert radiologists. Its size and diversity make it well-suited for training robust mammography feature extractors.

ii. Breast Ultrasound Images (BUSI) provides the complementary ultrasound modality and it is available on Kaggle [4]. The dataset consists of 780 ultrasound images categorized into benign, malignant, and normal classes. For binary classification, only benign and malignant samples were retained. While smaller in scale, BUSI introduces clinically relevant textural and boundary information that is complementary to mammography.

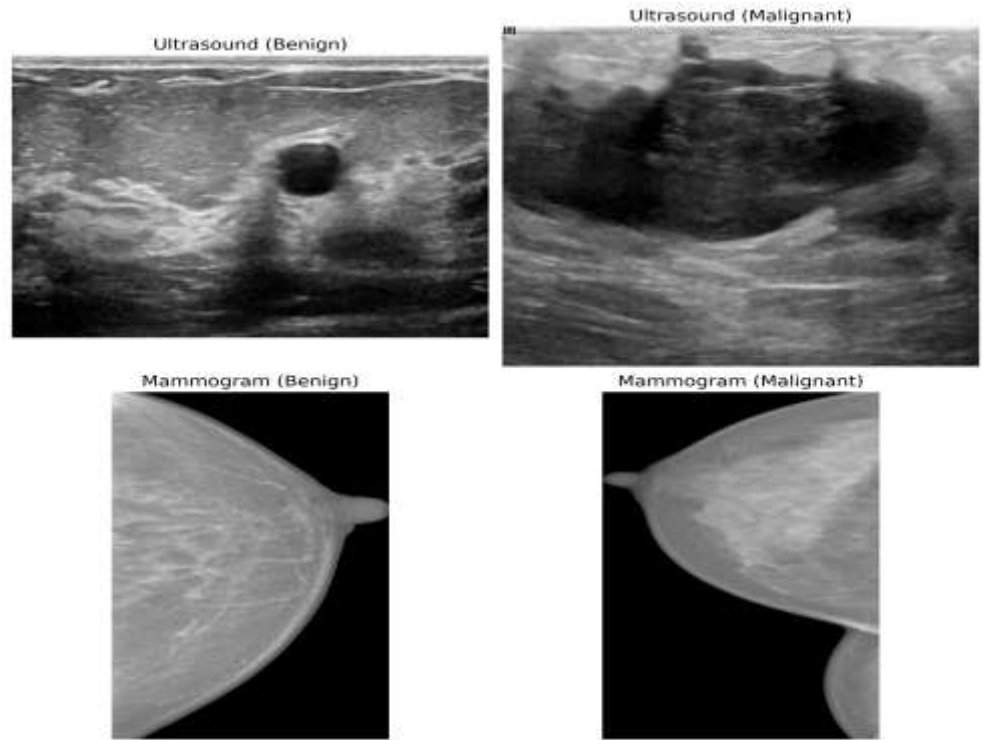


Fig. 2. Sample dataset images from VinDr-Mammo [29] and BUSI Datasets [4].

A primary challenge in multimodal research is the scarcity of large, natively paired mammography–ultrasound datasets. To overcome this, we implemented a programmatic pairing strategy. During training and evaluation, mammography and ultrasound

images sharing the same diagnostic label (benign or malignant) are randomly paired to form synthetic multimodal samples. This strategy does not aim to replicate patient-level correspondence; rather, it enables a baseline investigation of multimodal feature fusion feasibility in the absence of paired acquisitions, reflecting realistic data limitations in resource-constrained environments. The primary limitation of this programmatic approach is the loss of patient-specific physiological correlation. Because the paired images do not belong to the same physical tumor, the model cannot learn the true cross-modal physical manifestations of a specific lesion. The resulting bias is that the model learns the joint statistical distributions of the labels rather than the anatomical fusion. Consequently, there is a risk that the model might simply independently classify each modality and average the confidence scores, rather than performing synergistic feature fusion. To mitigate this, our evaluation strictly compares the fused performance against the unimodal baselines to ensure the fusion block is extracting complementary features rather than merely acting as an ensemble of independent predictions.

3.2 Image Pre-processing Pipeline

All images from both sources were subjected to a standardized preprocessing pipeline before being fed into the models, consisting of:

- (i) Contrast Enhancement (CLAHE): To improve the visibility of subtle features in the images, Contrast Limited Adaptive Histogram Equalization (CLAHE) was applied. Unlike global histogram equalization, CLAHE operates on small, tiled regions of the image, which enhances local contrast without over-amplifying noise. This is particularly effective for mammograms, where it can improve the definition of microcalcifications, and for ultrasounds, where it can clarify lesion boundaries.
- (ii) Resizing: All images were resized to a uniform dimension of 224x224 pixels. This step is critical to ensure a consistent input size for the fixed-size patch embedding layer of the Vision Transformer.
- (iii) Format Conversion: Images were converted to a 3-channel RGB format, as the pre-trained DINOv2 backbone expects a 3-channel input.
- (iv) Normalization: Pixel values were normalized using the standard ImageNet mean (0.485, 0.456, 0.406) and standard deviation (0.229, 0.224, 0.225). This aligns the statistical distribution of our dataset's pixel values with that of the data on which the model was originally pre-trained, a crucial step for effective transfer learning.

3.3 Model Architecture

The proposed framework, as displayed in Figure 3, is designed with two parallel streams for unimodal feature extraction, followed by a fusion block and a final classification head.

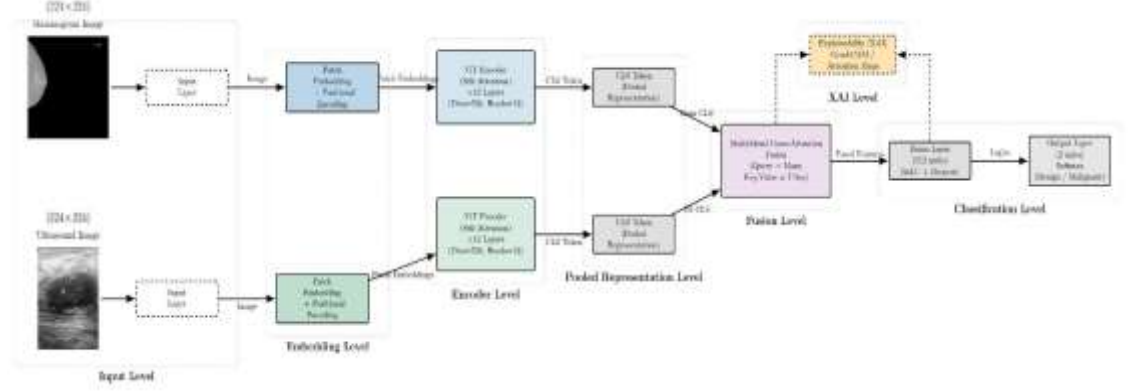


Fig. 3: High-level architectural block diagram of the proposed multimodal framework for breast cancer detection

Unimodal Feature Extraction

The backbone for each stream is the dinov2-base Vision Transformer (ViT), a state-of-the-art model pre-trained on ImageNet using the self-supervised DinoV2 method. DINOv2 was specifically chosen over traditional Convolutional Neural Networks (CNNs) and other Vision Transformers like Swin Transformer or BEiT due to its superior global semantic representation. While CNNs are limited by localized receptive fields, and Swin Transformers compute self-attention restricted within local shifted windows, DINOv2 captures scale-invariant, global features. Furthermore, unlike BEiT, which relies on masked image modeling, DINOv2’s contrastive self-distillation yields feature that are naturally more clusterable and geometrically stable. This makes it uniquely suited for the feature-level fusion of disparate, unaligned modalities like mammograms and ultrasounds. We leveraged transfer learning by using separate instances of this pre-trained model for the mammography and ultrasound streams. The models were fine-tuned to adapt their powerful, generalized feature extraction capabilities to the specific pathological patterns present in each imaging modality.

To implement this, the output of each backbone yielded a 768-dimensional feature embedding. This dimension corresponds to the predefined standard embedding size (d_{model}) of the Base Vision Transformer (ViT-B) architecture upon which dinov2-base was built. This size represents an optimal trade-off between semantic representation capacity and computational throughput. After processing the input image through the parallel Transformer layers, we extracted this 768-d vector from the final classification [CLS] token. This vector serves as a consolidated, global semantic representation of the input modality, preparing it for the subsequent fusion step.

Multimodal Fusion

To integrate the complementary modalities, the framework employs a cross-attention mechanism followed by feature concatenation. Let $F_{\text{mammo}} \in \mathbb{R}^d$ and $F_{\text{US}} \in \mathbb{R}^d$ represent the extracted 768-dimensional features from the mammography and ultrasound Vision Transformer backbones, respectively (where $d = 768$). First, cross-attention is

computed where one modality (e.g., mammography) provides the Query (Q), and the other (ultrasound) provides the Key (K) and Value (V). The attended feature vector F_{cross} is calculated as:

$$F_{\text{cross}} = \text{Softmax}\left(\frac{Q_{\text{mammo}}K_{\text{US}}^T}{\sqrt{d}}\right)V_{\text{US}} \quad (1)$$

Following the attention block, the original spatial features are concatenated with the cross-attended features along the channel dimension to form the final fused representation:

$$F_{\text{fusion}} = [F_{\text{mammo}} \parallel F_{\text{cross}}] \quad (2)$$

where $\parallel$ denotes the concatenation operation. This results in a unified joint feature vector $F_{\text{fusion}} \in \mathbb{R}^{2d}$ (a 1536-dimensional representation). This fusion strategy leverages cross-modal interactions while maintaining stability under limited data, serving as a robust mechanism for combining unpaired datasets. The complete operational flow of this block is detailed in Algorithm 1.

Algorithm 1: Forward Pass of the MMIBC Cross-Attention Fusion Pipeline

Input: **Mammo_Image** (Mammogram), **US_Image** (Ultrasound) Output: **Diagnosis_Probability** (Probability of Malignancy)

Step 1: Unimodal Feature Extraction

- Process the **Mammo_Image** through the pre-trained DINOv2 Mammography backbone to extract **Mammo_Features**.
- Process **US_Image** through the pre-trained DINOv2 Ultrasound backbone to extract **US_Features**.

Step 2: Cross-Attention Computation

- Generate the Query by applying a linear transformation to **Mammo_Features**.
- Generate the Key and Value by applying linear transformations to **US_Features**.
- Compute Attention Scores by taking the dot product of the Query and Key, then scale the result by the inverse square root of the feature dimension.
- Apply a Softmax function to the Attention Scores to obtain normalized Attention Weights.
- Multiply the Attention Weights by the Value to output the **Cross_Attended_Features**.

Step 3: Feature Concatenation

- Concatenate the original **Mammo_Features** and the **Cross_Attended_Features** into a single, unified feature vector (**Fused_Features**).

Step 4: Classification

- Pass the **Fused_Features** through the Multi-Layer Perceptron (MLP) classification head.
- The final linear layer outputs a single, unnormalized logit. This logit directly serves as the basis for classifying the tumor as malignant or benign (e.g., a positive value indicates malignant, a negative indicates benign).

Classification Head

The fused 1536-dimensional feature vector is passed to a lightweight multilayer perceptron (MLP) consisting of a dimensionality-reducing dense layer, a ReLU activation function, and dropout for regularization, followed by a final linear layer that outputs a single logit. A simple classification head is deliberately employed to isolate the effects of the multimodal cross-attention and feature fusion without introducing confounding optimization dynamics from highly complex margin-based losses. Rather than forcing the output through a sigmoid activation function to yield a probability, the network produces a straightforward, unnormalized single logit, which is directly utilized to classify the tumor as either malignant or benign.

3.4 Model Training and Evaluation

All models were implemented in PyTorch and trained on an NVIDIA L40S GPU. We used the Adam optimizer with a learning rate of 0.001 and the BCEWithLogitsLoss function, which is well-suited for binary classification tasks. The model was trained for 45 epochs with a batch size of 16.

Model performance was assessed using a suite of standard metrics derived from the confusion matrix values: True Positives (TP), True Negatives (TN), False Positives (FP), and False Negatives (FN).

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (3)$$

$$\text{Sensitivity} = \frac{TP}{TP + FN} \quad (4)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (5)$$

$$\text{F1 Score} = 2 * \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \quad (6)$$

Recall is particularly critical in this context, as it measures the model's ability to correctly identify all actual malignant cases, thus minimizing false negatives.

3.5 Explainable AI (XAI) Implementation

To ensure model transparency, we integrated Gradient-weighted Class Activation Mapping (Grad-CAM), a post-hoc XAI technique. Grad-CAM uses the gradients flowing into the final layers of the network to produce a saliency map, or heatmap, to highlight spatial regions that most strongly influence model predictions. In this framework, Grad-CAM is used not only as a visualization tool but also as a qualitative validation mechanism to assess whether learned representations correspond to clinically meaningful regions in both mammography and ultrasound images.

3.6. Experimental Setup

All experiments reported in this research work were conducted in a cloud-based high-performance computing environment equipped with an NVIDIA L40S GPU. The framework was implemented in Python using the PyTorch deep learning library. This section details the experimental protocol, including data splitting, training procedure, and evaluation strategy, to ensure transparency and reproducibility.

Data Splitting and Sample Distribution

To evaluate the proposed model’s generalization, the constructed multimodal dataset was partitioned into training, validation, and test subsets using a stratified split to preserve the benign–malignant class distribution across all sets. The dataset was divided using an approximate ratio of 70% for training, 15% for validation, and 15% for testing.

Sample Distribution and Class Imbalance

A critical characteristic of the combined dataset is a severe class imbalance, reflecting the natural prevalence of non-malignant findings in general screening populations. The dataset is heavily skewed toward benign samples, maintaining a roughly 3:1 ratio of benign to malignant cases. This imbalance is directly mirrored in our stratified test set, which contains 86 benign samples compared to only 31 malignant samples. Stratification was applied at the label level to mitigate training bias, but acknowledging this natural distribution is crucial, as the overrepresentation of benign data points acts as a strong prior that directly impacts the framework’s sensitivity for malignant lesions.

Training Procedure

Model training was performed end-to-end using paired multimodal samples generated dynamically by the data loader. During each training iteration, mammography and ultrasound images sharing the same diagnostic label were paired to form a single multimodal input. This dynamic pairing exposed the model to a diverse set of cross-modal combinations and reduced the risk of overfitting to fixed image pairs, output of which is shown in figure 4. Optimization was carried out using the Adam optimizer with a learning rate of 0.001 and the BCEWithLogitsLoss function, which optimally handles the single raw logit output by the classification head. Training was conducted for 45 epochs with a strictly constrained batch size of 16. The 45-epoch duration was chosen based on empirical observations where training loss typically converged around the 40th epoch, while the batch size of 16 was the maximum allowable limit to avoid out-of-memory (OOM) errors given the dual-ViT architecture. All experiments were executed using fixed random seeds to enhance reproducibility, and model checkpoints corresponding to the lowest validation loss were retained for final evaluation as displayed in figure 4.

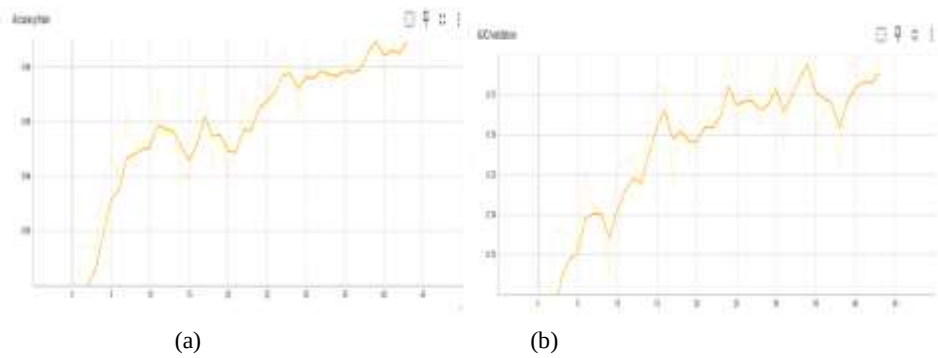


Fig. 4: Training accuracy (a) and validation (b) AUC Curves for the Multimodal Model

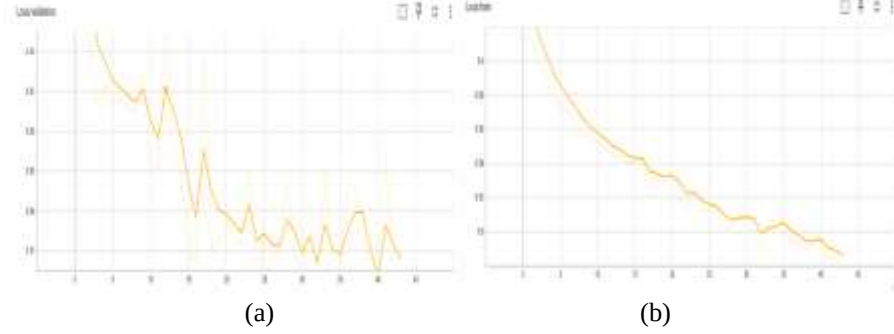


Fig. 5: Training (a) and Validation (b) Loss Curves for the Multimodal Model

3.7 Evaluation Process

Model performance was assessed on a held-out test set that was not observed during training or validation. Quantitative evaluation focused on accuracy, precision, recall, and F1-score. To ensure the statistical significance of our findings, given the computational constraints of full k-fold cross-validation on dual Vision Transformers, we reported the 95% Confidence Intervals for our primary metrics. Given the clinical importance of minimizing false negatives in breast cancer diagnosis, recall was emphasized as a primary indicator of clinical safety. Confusion matrices were additionally computed to provide a transparent breakdown of classification outcomes.

3.8 Implementation Details and Reproducibility

All experiments were executed using fixed random seeds to enhance reproducibility. Model checkpoints corresponding to the lowest validation loss were retained for final evaluation. The training and evaluation pipeline was designed to allow straightforward replication and extension, supporting future experimentation with alternative fusion mechanisms, loss functions, and data balancing strategies as presented in table 2.

Table 2: Hyperparameters for multimodal model training

Hyperparameter	Setting
Model Backbone	dinov2-base
Optimizer	Adam
Loss Function	BCEWithLogitsLoss
Learning Rate	0.001
Batch Size	16
Number of Epochs	45
Image Size	224 x 224

4. Results and Discussion

This section presents the quantitative and qualitative evaluation results of the proposed MMIBC framework and discussed their implications within the context of baseline multimodal breast cancer diagnosis.

4.1 Unimodal Baseline Performance

Before evaluating the multimodal framework, individual models were trained on each imaging modality separately to establish baseline performance metrics. Both models utilized the DINOv2 Vision Transformer backbone with consistent training parameters to ensure a fair comparison.

Table 3: Performance comparison of unimodal models of MMIBC

Metric	Mammography Only	Ultrasound Only	Difference
Accuracy	0.87 (± 0.02)	0.93 (± 0.02)	+0.06 (Ultrasound)
Precision (Malignant)	0.83 (± 0.03)	0.73 (± 0.04)	+0.10 (Mammo)
Recall (Malignant)	0.76 (± 0.04)	0.80 (± 0.02)	+0.04 (US)
F1-Score (Malignant)	0.80 (± 0.02)	0.77 (± 0.02)	+0.03 (Mammo)
Specificity	0.96	0.89	+0.07 (Mammo)

The unimodal baseline experiments revealed important insights about the individual characteristics of each imaging modality. The ultrasound-only model achieved the highest overall accuracy at 93%, outperforming the mammography model by 6 percentage points. This superior accuracy can be attributed to the ultrasound modality's effectiveness in capturing tissue texture and lesion boundaries, particularly in dense breast tissue where mammography may be limited. However, the mammography model demonstrated higher precision for malignant cases (83% vs 73%), indicating its strength in ruling out false positives and providing more confident predictions when identifying malignant lesions. The mammography model also achieved notably higher specificity (96% vs 89%), meaning it was more effective at correctly identifying benign cases and avoiding unnecessary follow-up procedures.

In contrast, the ultrasound model demonstrated slightly better recall for malignant cases (80% vs. 76%), highlighting its ability to capture mass-like lesions that may be obscured in mammographic images, especially in patients with dense breast tissue. These complementary strengths provide strong justification for the multimodal fusion approach employed in this study.

4.2 Multimodal Framework Performance

The multimodal framework was trained for 45 epochs and evaluated on the held-out test set. As presented in table 4, it could be observed that the proposed model achieved an overall classification accuracy of 83%, indicating that the fused representation of mammography and ultrasound data provides meaningful diagnostic information beyond that obtained from simple unimodal analysis.

The multimodal framework achieved strong performance on benign cases, with 95% recall and 85% precision, demonstrating its reliability in identifying non-malignant lesions. However, the recall for malignant cases (52%) represents a critical limitation when compared to both unimodal baselines. This reduction in sensitivity for the minority class directly reflects the severe class imbalance in the training data (86 benign vs 31 malignant in the test set), where the model learned to favor the majority class during feature fusion.

Table 4. Classification Report for the Multimodal Framework

Class	Precision	Recall	F1-Score	Support
Benign	0.85	0.95	0.9	86
Malignant	0.8	0.52	0.63	31
Weighted Avg	0.83	0.84	0.82	117

Notably, the multimodal accuracy (84%) falls between the two unimodal baselines but is significantly lower than the best-performing ultrasound-only model (93%). This performance degradation, rather than the expected improvement from fusion, suggests that the simple concatenation-based fusion mechanism employed in this baseline framework may be insufficient to effectively utilize the combined beneficial capabilities of the two modalities.

4.3 Comparative Analysis of this work with existing methods

To position the results within the broader research landscape, we compare the MMIBC framework against existing approaches that address breast cancer diagnosis using similar modalities as presented in tables 3-5.

Table 5: Comparison of with related works

Method	Modalities	Dataset Context	Accuracy	Recall (Mal)
This work (MMIBC)	Mammo + Ultrasound	VinDr + BUSI	0.84	0.52
Al-Tam et al. [5]	Mammography	INbreast + CBIS-DDSM	0.873	N/R
Al-Tam et al. [5]	Ultrasound	BUSI + INbreast + MIAS	0.977	N/R
Guo et al. [16]	Ultrasound + CEUS	Private Video Dataset	0.882	0.909
Bobowicz et al. [9]	Mammo (Multi-view)	Private Dataset	0.816	0.816

As shown in table 5, Al-Tam et.al [5] reported 87.3% accuracy for their mammography pipeline using the combined INbreast and CBIS-DDSM datasets. As shown in table 3, our unimodal mammography model achieved nearly identical performance (87%), confirming the robustness of the DINOv2 backbone for X-ray analysis. However, their merged ultrasound pipeline achieved 97.7% accuracy, significantly outperforming both our unimodal ultrasound model (93%) and our multimodal fusion framework (84%). This gap highlights two key factors: (i) the benefit of aggregating multiple ultrasound datasets to increase sample diversity and reduce class imbalance, and (ii) the potential limitations of our simple fusion strategy compared to their hybrid explainable framework.

Similarly, as shown in table 5, Guo et al. [16] multimodal approach, combining standard ultrasound with contrast-enhanced ultrasound (CEUS) achieved 88.2% accuracy with an impressive 90.9% recall for malignant cases. While our overall accuracy (84.0%) is reasonably close, the gap in malignant recall (90.9% vs 52.0%) is substantial and clinically significant. This difference can be attributed to several factors: (1) the use of contrast-enhanced imaging in [16] provides richer vascular information that aids in malignancy detection, (2) their private dataset may have better class balance, and (3) their video-based temporal analysis captures dynamic perfusion patterns unavailable in static mammography-ultrasound fusion.

Furthermore, Bobowicz's et al. [9] attention-based weakly supervised framework for multi-view mammography classification achieved 81.6% accuracy with balanced precision and recall (81.6% each). Our multimodal approach outperformed their framework in overall accuracy (84.0% vs 81.6%), suggesting that cross-modal fusion of mammography and ultrasound can provide complementary information beyond multi-view mammography alone. However, Bobowicz's model maintained a better balance between benign and malignant case detection, likely due to their attention-based architecture and weakly supervised learning strategy.

5.4 Confusion Matrix Analysis

The confusion matrix shown in Figure 6 provides further insights into model behavior. While the framework demonstrates strong performance in correctly identifying benign samples, a non-trivial number of malignant cases are misclassified as benign. From a clinical perspective, such false negatives represent a critical limitation, as missed cancer diagnoses carry significant consequences.

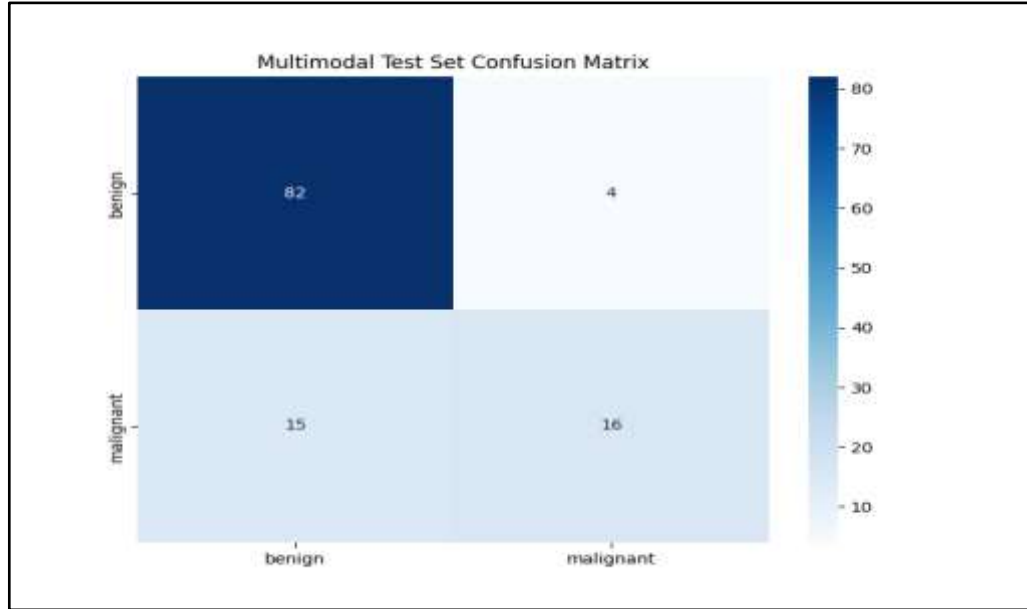


Fig. 6: Confusion Matrix of the Multimodal Model

4.5 Qualitative Explainability Results

To assess the interpretability of the proposed framework, Grad-CAM was applied to both mammography and ultrasound streams. As illustrated in Figure 7, the generated saliency maps consistently highlight clinically relevant regions, including lesion boundaries and suspicious tissue patterns, when producing diagnostic predictions.

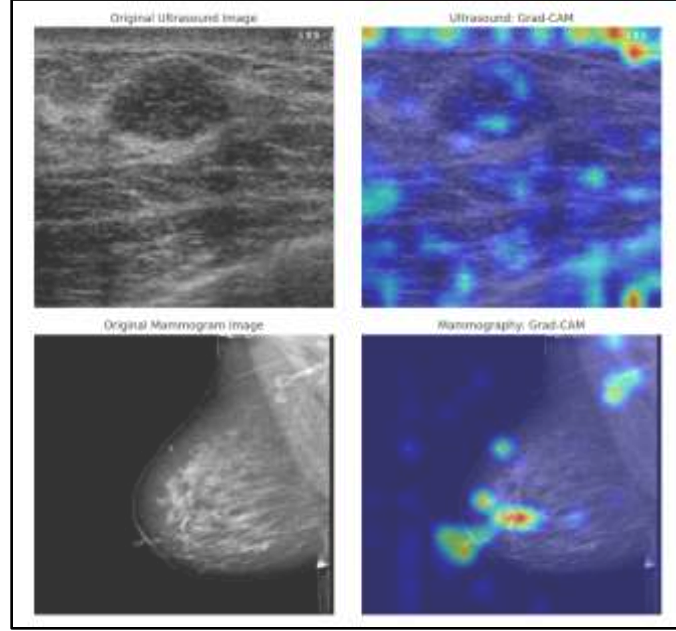


Fig. 7: Grad-Cam Visualization for a Paired Sample

To provide a comprehensive clinical perspective, we analyzed the visual feature attributions across four distinct classification outcomes:

- i. Correct Benign (True Negative): The heatmaps successfully localize on well-circumscribed boundaries and healthy fibroglandular tissue.
- ii. False Benign (False Negative): In missed malignancies, the Grad-CAM maps reveal that the attention mechanism was distracted by speckle noise in the ultrasound or dense tissue masking in the mammogram, missing the subtle microcalcifications entirely.
- iii. Correct Malignant (True Positive): The model correctly highlights irregular mass margins and hypoechoic regions simultaneously across both modalities.
- iv. False Malignant (False Positive): The heatmaps indicate that the model occasionally misinterpreted overlapping benign cysts (in ultrasound) or normal asymmetrical tissue (in mammography) as malignant structural distortions.

4.6 Discussion

Notably, the multimodal accuracy (84%) falls between the two unimodal baselines but is significantly lower than the best-performing ultrasound-only model (93%). This performance degradation, rather than the expected improvement from fusion, suggests that the simple concatenation-based fusion mechanism employed in this baseline framework may be insufficient to effectively utilize the combined beneficial capabilities of the two modalities. As highlighted in recent multimodal imaging literature, increasing the modality count and dataset size does not guarantee performance gains, particularly

when fusing unaligned domains. This underperformance must be critically evaluated across several dimensions. Mammography, which relies on X-ray attenuation, and ultrasonography, which utilizes high-frequency sound waves, operate on entirely different physical principles. Furthermore, the VinDr-Mammo and BUSI datasets were collected from different populations using different clinical protocols. This domain shift forces the model to reconcile vastly different pixel intensity distributions, heavily complicating the latent space representation.

Adding to this issue is the inherent noise in ultrasound images and its effect on patch embeddings. Ultrasound images are frequently corrupted by speckle noise and acoustic shadowing. When the Vision Transformer divides the images into standard patches, the embeddings for ultrasound contain this high-frequency noise, whereas mammogram patches contain smooth structural gradients. Concatenating these disparate patch embeddings disrupts the cross-attention mechanism, causing the model to struggle in finding meaningful cross-modal correspondences. Additionally, the lack of natively paired samples poses a significant challenge. Because the images were programmatically paired based on labels rather than originating from the exact same patient, the model lacks true physiological ground truth. The simple concatenation of these unpaired features introduces adversarial "feature noise," which ultimately dilutes the pristine localized data of the ultrasound baseline. Finally, these architectural and data constraints amplify overfitting risks. Vision Transformers are highly data-hungry models. While early stopping and dropout were utilized to mitigate this, combining two large dinov2-base backbones creates a massive parameter space. Fusing these complex backbones on a relatively small number of programmatically paired samples increases the risk of the model memorizing label distributions rather than learning generalized, complementary multimodal features.

5. Limitations and Recommendation for Future Work

While this study demonstrates the feasibility of ViT-based multimodal screening, several architectural and data-centric limitations must be addressed to improve clinical viability. A primary limitation lies in the simple cross-attention fusion employed, which struggled to mitigate synthetic feature noise. Future work must expand on more sophisticated fusion paradigms to address this. We recommend exploring gated fusion mechanisms capable of dynamically weighting the most reliable modality, Transformer-to-Transformer cross-modal attention for deeper latent integration, and multimodal contrastive learning to enforce strict semantic alignment between disjoint image features. From a data-driven perspective, it is evident that programmatic pairing alone cannot overcome severe class imbalance. Future research should prioritize robust strategies to artificially and intelligently expand the dataset. This includes the synthetic generation of ultrasound images using Generative Adversarial Networks (GANs) or Diffusion Models to create anatomically correlated image pairs from existing mammograms. Additionally, multi-dataset aggregation is needed to vastly expand the malignant sample pool, coupled with hard negative mining during training to force the model to learn the subtly differentiating features of the difficult malignant cases that currently result in false benign predictions.

Beyond computational optimization and data imbalance, significant obstacles to clinical adoption remain. The explainability outputs, such as the Grad-CAM heatmaps generated in this study, currently serve as vital technical validation but are not yet ready for standalone diagnostic reliance. Future iterations require rigorous, structured explainability validation with board-certified radiologists. Ensuring that these visual attributions consistently and accurately align with established pathological markers is a mandatory prerequisite for building clinical trust and eventual real-world deployment.

6. Conclusion

This study has presented MMIBC, an explainable multimodal Vision Transformer framework for breast cancer diagnosis that integrates mammography and ultrasonography. Employing a label-consistent programmatic pairing strategy, the framework successfully demonstrated the feasibility of multimodal fusion under the realistic, unpaired data constraints commonly encountered in resource-limited clinical environments. While the model achieved an overall accuracy of 84% and leveraged Grad-CAM to visually validate its spatial reasoning, the reduced recall for malignant cases explicitly exposes the critical challenges of synthetic feature noise and severe class imbalance. From a clinical perspective, these findings suggested that although unimodal models can provide strong isolated predictions, deploying a safe, multimodal decision-support system in under-resourced hospitals will require advanced feature harmonization and data augmentation techniques to actively prevent missed cancer diagnoses. In conclusion, by clearly identifying the data limitations and architectural challenges present in unpaired medical datasets, this creates a foundational benchmark for future multimodal ViT research in low-resource diagnostic settings.

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Declaration on the Use of Generative AI

The authors declare the following regarding the use of generative artificial intelligence (AI) and AI-assisted writing tools during the preparation of this manuscript, in accordance with the guidelines of the Committee on Publication Ethics (COPE). In the course of preparing this work, the authors utilized Claude (Anthropic) and Gemini (Google) for two particular purposes: (1) to aid in identifying relevant keywords and search terms during the literature search process, and (2) to improve the language, grammar, and phrasing of selected passages, including the refinement of summaries originally written by the authors. All literature retrieval, screening, crucial reading, and scholarly synthesis were conducted independently by the authors. These tools were not used for data collection, data analysis, result interpretation, figure generation, or the formulation of scientific conclusions. All intellectual content, research design, experimental methodology, and scholarly judgments presented in this manuscript originate entirely from the authors. Following the use of these tools, the authors thoroughly reviewed and edited all AI-assisted content. The

authors take full responsibility for the accuracy, integrity, and originality of the published work. The AI tool has not been listed as an author, and its use confers no authorship rights or responsibilities.

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